

Example - Electric Dipole & Atmospheric Sprites:

Given Data:

$$q = 200 \text{ C}$$

$$h = 6 \text{ km} = 6 \times 10^3 \text{ m}$$

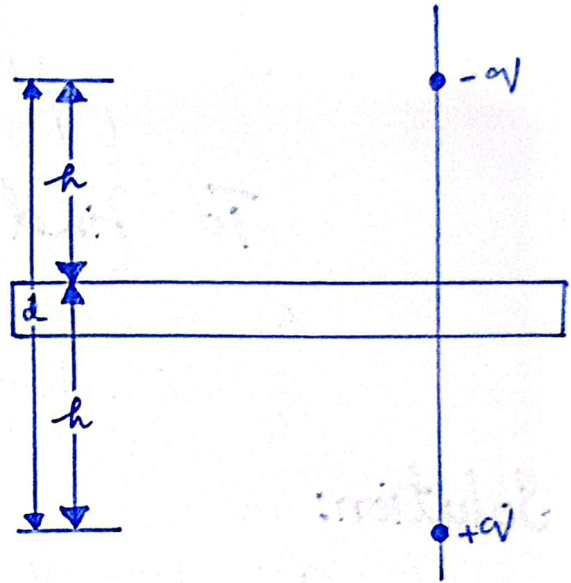
$$z_1 = 30 \text{ km} = 30 \times 10^3 \text{ m}$$

$$z_2 = 60 \text{ km} = 60 \times 10^3 \text{ m}$$

To find

$$E_1 = ?$$

$$E_2 = ?$$



Solution:

$$E = \frac{1}{2\pi\epsilon_0} \frac{q d}{z^3}$$

so For z_1 : $(\because d = 2h)$

$$\begin{aligned} E_1 &= \frac{1}{2\pi\epsilon_0} \frac{q(2h)}{z_1^3} \\ &= \frac{1}{2\pi\epsilon_0} \frac{200 \times 2 \times 6 \times 10^3}{(30 \times 10^3)^3} \end{aligned}$$

For z_2 : $\boxed{= 1.6 \times 10^3 \text{ N/C}}$

$$\begin{aligned} E_2 &= \frac{1}{2\pi\epsilon_0} \frac{q(2h)}{z_2^3} \\ &= \frac{1}{2\pi\epsilon_0} \frac{(200)(2)(6 \times 10^3)}{(60 \times 10^3)^3} \end{aligned}$$

$$\boxed{= 2 \times 10^2 \text{ N/C}}$$

Problems

Q.3:-

Given Data

$$n = Z = 94 \text{ protons.}$$

$$r = 6.64 \text{ fm} = 6.64 \times 10^{-15} \text{ m}$$

To find

a) $E = ?$

b) Direction = ?

Solution:

$$a) \quad E = \frac{q}{4\pi\epsilon_0 r^2} = \frac{94 \times 1.6 \times 10^{-19} \times 9 \times 10^9}{(6.64 \times 10^{-15})^2}$$

$$E = 3.1 \times 10^{21} \text{ N/C}$$

b) Direction of the Electric field to the surface will be outward because of the positive charge.

Q.5:-

Given Data

$$E = 2 \text{ N/C}, \quad r = 50 \text{ cm} = 50 \times 10^{-2} \text{ m}$$

To find

$$|q| = ?$$

Solution:

$$E = \frac{|q|}{4\pi\epsilon_0 r^2} \Rightarrow |q| = 4\pi\epsilon_0 r^2 E = 5.6 \times 10^{-11} \text{ C}$$

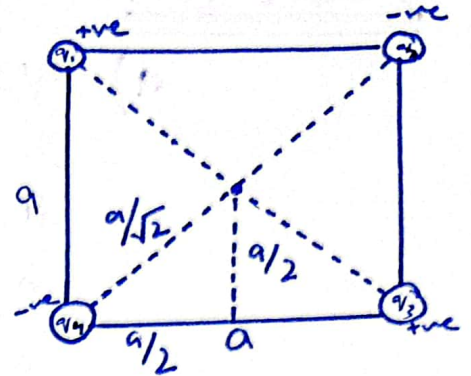
7:

Given Data

$$\begin{aligned} q_1 &= +10 \text{ nC} \\ q_2 &= -20 \text{ nC} \\ q_3 &= +20 \text{ nC} \\ q_4 &= -10 \text{ nC} \\ a &= 5 \text{ cm} = 5 \times 10^{-2} \text{ m} \end{aligned}$$

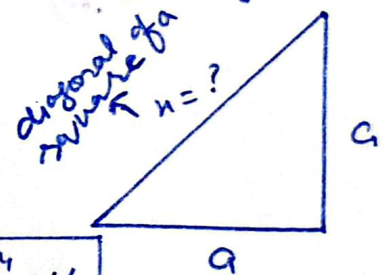
To find

$$E_{\text{net}} = ?$$



Solution:

$$\begin{aligned} E_x &= \frac{1}{4\pi\epsilon_0} \left(\frac{|q_2|}{(a/\sqrt{2})^2} - \frac{|q_4|}{(a/\sqrt{2})^2} \right) \rightarrow \text{x-axis } q_2 \text{ \& } q_4 \text{ (half of diagonal)} \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{20}{a^2/2} - \frac{10}{a^2/2} \right) \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{10}{a^2/2} \right) \\ &= \frac{9 \times 10^9 \times 10 \times 10^{-9}}{(5 \times 10^{-2})^2 / 2} = \boxed{7.19 \times 10^4 \text{ N/C}} \end{aligned}$$



Similarly

$$E_y = 7.19 \times 10^4 \text{ N/C} \rightarrow \text{y-axis } q_1 \text{ \& } q_3$$

So,

$$\begin{aligned} E &= \sqrt{E_x^2 + E_y^2} \\ &= \sqrt{(7.19 \times 10^4)^2 + (7.19 \times 10^4)^2} \end{aligned}$$

$$\boxed{E = 1.02 \times 10^5 \text{ N/C}}$$

Q.22:-

a) $q = -300e$

$$r = 4 \text{ cm} = 4 \times 10^{-2} \text{ m}$$

$$\theta = 40^\circ$$

$$\lambda = ?$$

$$\lambda = \frac{qV}{L}$$

$$S = r \theta$$

$$S = L = 0.0279 \text{ m}$$

$$q = -300 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-17} \text{ C}$$

$$\lambda = \frac{4.8 \times 10^{-17}}{0.0279} = 1.72 \times 10^{-15} \text{ C/m}$$

b)

$$q = -300e$$

$$r = 2 \text{ cm}$$

$$\sigma = \frac{qV}{A}$$

$$A = \pi r^2 = 3.14 (2 \times 10^{-2})^2 \\ = 1.256 \times 10^{-3} \text{ m}^2$$

$$\sigma = \frac{-300 \times 1.6 \times 10^{-19}}{1.256 \times 10^{-3}} \Rightarrow \sigma = -3.82 \times 10^{-4} \text{ C/m}^2$$

c)

$$q = -300e$$

$$r = 2 \text{ cm}$$

$$A = 4\pi r^2$$

$$= 4 \times 3.14 (2 \times 10^{-2})^2$$

$$\sigma = \frac{qV}{A} = \frac{-300 \times 1.6 \times 10^{-19}}{4\pi r^2}$$

$$\sigma = -9.5 \times 10^{-15} \text{ C/m}^2$$

d)

$$q = -300e$$

$$r = 2 \times 10^{-2} \text{ m}$$

$$\rho = \frac{q}{\text{Vol.}}$$

$$\rho = \frac{-300 \times 1.6 \times 10^{-19}}{\frac{4}{3} \pi r^3} = \frac{-4.8 \times 10^{-17}}{\frac{4}{3} \times 3.14 (2 \times 10^{-2})^3}$$

$$\boxed{\rho = -1.43 \times 10^{-12} \text{ C/m}^3}$$

Q. 42:-

$$E = 3 \times 10^6 \text{ N/C}$$

a) $F = ?$ on e^-

b) $F = ?$ on ion

a) $F_e = eE = 1.6 \times 10^{-19} \times 3 \times 10^6 = 4.8 \times 10^{-13} \text{ N}$

b) $F_i = Eq = 3 \times 10^6 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-13} \text{ N}$

Q. 43:-

$$E = 2 \times 10^4 \text{ N/C}$$

acc. of the electron = ?

$$a = \frac{F}{m} = \frac{eE}{m} = \frac{1.6 \times 10^{-19} \times 2 \times 10^4}{9.11 \times 10^{-31}}$$

$$\boxed{a = 3.51 \times 10^{15} \text{ m/s}^2}$$

Q. 44:-

$$F_E = F_g$$

For α -particle $qE = mg \Rightarrow E = \frac{mg}{q}$

$$E = \frac{mg}{2e} = 2 \times 10^7 \text{ N/C}$$

Q.47:

a) acceleration of proton = ?

$$E = 2 \times 10^4 \text{ N/C}$$

b) speed of proton = ?

through a distance of 1 cm.

$$a = \frac{F}{m} = \frac{eE}{m} = \frac{1.6 \times 10^{-19} \times 2 \times 10^4}{1.67 \times 10^{-27}}$$

$$a = 1.9 \times 10^{12} \text{ m/s}^2$$

b) Suppose proton was at rest initially, so

$$v^2 = v_0^2 + 2ad$$

$$v_0 = 0$$

so,

$$v = \sqrt{2ad}$$

$$v = \sqrt{2 \times 1.9 \times 10^{12} \times 1 \times 10^{-2}}$$

$$v = 1.96 \times 10^5 \text{ m/s}$$